

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 3: Multi-Engine Class B Piston (Take-Off & OEI Climb)

Syllabus Source: CAE Oxford (pp. 79-114) | Pass Target: $\geq 90\%$ AviationExam

1. ASYMMETRIC FLIGHT & LOSS OF 80% CLIMB PERFORMANCE

When an engine fails on a twin-engine piston aircraft, 50% of the total engine power is lost. **HOWEVER, CLIMB PERFORMANCE DROPS BY 80% OR MORE!** This is because climb depends on **EXCESS POWER** above that required for level flight:

- **Excess Power Law:** Power Available drops by 50%, while Power Required **INCREASES** due to rudder deflection drag and windmilling propeller drag. The remaining margin of excess power is cut by 80% to 100%!
- **Windmilling Drag:** An unfeathered windmilling propeller produces massive drag (equivalent to dragging a large flat plate through the air). **IMMEDIATE ACTION:** Feather the dead propeller!
- **5° Bank Rule:** Banking 2° to 5° **TOWARDS THE OPERATING ENGINE** and centering the ball with half-deflection creates a horizontal component of lift that opposes engine yaw, reducing rudder drag and maximizing OEI climb gradient.

2. MULTI-ENGINE SERVICE CEILING & SPEEDS

- **OEI Service Ceiling:** The altitude where the single-engine rate of climb drops to 50 ft/min (with propeller feathered).
- **Vyse (Blue Line Speed):** Best Rate of Climb Speed with One Engine Inoperative. Marked on the airspeed indicator by a prominent **BLUE RADIAL LINE**. The single most important safety speed in twin-engine emergencies!
- **Vmca (Red Line Speed):** Minimum Control Speed in the Air. Marked by a **RED RADIAL LINE**. Airspeed **MUST NEVER** be allowed to drop below Vmca during asymmetric flight!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Multi-Engine Class B Engine Failure

Step 1: Directional Control: Apply full rudder toward live engine and bank up to 5 degrees into live engine. Step 2: Maintain Airspeed: Pitch to hold VYSE (Blue Line on airspeed indicator). Step 3: Cleanup Airframe: Landing gear UP, Flaps UP. Step 4: Identify & Feather: Dead foot, dead engine. Feather inoperative propeller immediately! Rule: If airspeed drops below VMCA (Red Line), reduce live engine throttle immediately to prevent unrecoverable roll!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Determining OEI Drift Down & Terrain Clearance

SCENARIO (Twin-Engine OEI Emergency): A light twin-engine aircraft is cruising at 9,000 ft MSL: - Aircraft All-Engine Service Ceiling: 18,000 ft - Certified One-Engine Inoperative (OEI) Service Ceiling at current mass: 6,500 ft MSL - The left engine fails catastrophically! - Flight path crosses a mountain ridge with elevation 6,800 ft MSL located 20 miles ahead QUESTION: Can the aircraft maintain level flight at 9,000 ft? What will happen, and can it clear the ridge? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check if 9,000 ft can be sustained: - OEI Service Ceiling = 6,500 ft MSL. - Because current altitude (9,000 ft) is HIGHER than the OEI ceiling (6,500 ft), the aircraft CANNOT sustain level flight! - The aircraft MUST descend (drift down) toward 6,500 ft. Step 2: Compare OEI ceiling with the mountain ridge: - OEI Service Ceiling = 6,500 ft MSL. - Mountain ridge height = 6,800 ft MSL. - Since 6,500 ft < 6,800 ft, the aircraft will eventually settle BELOW the top of the mountain ridge! Step 3: Operational Decision / Emergency Procedure: - Setting blue line speed (Vyse) will only delay descent down to 6,500 ft. - The pilot MUST IMMEDIATELY TURN AWAY from the mountain ridge toward lower terrain or an emergency diversion airport! FINAL ANSWER: Flight at 9,000 ft cannot be maintained. Aircraft will drift down to 6,500 ft, which is below the 6,800 ft ridge. Immediate diversion is mandatory.

[!] EXAM TRAP (AVIATIONEXAM) - The 50% Power vs 80% Climb Trap

- Losing 1 engine on a twin LOSES 50% OF POWER, BUT 80% TO 90% OF CLIMB RATE! - Example: Twin climbing at 1,000 ft/min on 2 engines does NOT climb at 500 ft/min on 1 engine! It will climb at only 100 to 200 ft/min!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: On a multi-engine piston aircraft airspeed indicator, what does the BLUE RADIAL LINE indicate?**

- [A]: Maximum flap extension speed
- [B]: Vyse (Best Rate of Climb Speed with One Engine Inoperative)
- [C]: Minimum control speed ground
- [D]: Never exceed speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The blue line marks Vyse, the airspeed yielding maximum single-engine rate of climb with the critical engine feathered.

Q2: Why does an aircraft lose approximately 80% of its climb performance when one engine fails on a twin-engine aircraft?

- [A]: Half the fuel is lost
- [B]: Climb depends on EXCESS power; after losing 50% power and adding asymmetric drag, the remaining excess power above level flight is tiny
- [C]: The tailplane stalls
- [D]: Generator fails

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Climb requires excess thrust above total drag. At single-engine speed, almost all remaining thrust is consumed just keeping the airplane in level flight, leaving very little excess power for climb.

Q3: Under CS-23, what is the definition of OEI Service Ceiling for multi-engine piston aircraft?

- [A]: Climb rate = 0 ft/min
- [B]: Altitude where maximum rate of climb with one engine feathered drops to 50 ft/min
- [C]: 100 ft/min
- [D]: 15,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. For multi-engine Class B aircraft, OEI Service Ceiling is defined at a residual climb rate of 50 ft/min (compared to 100 ft/min for all-engine).